



Phishing Email Detection & Awareness System

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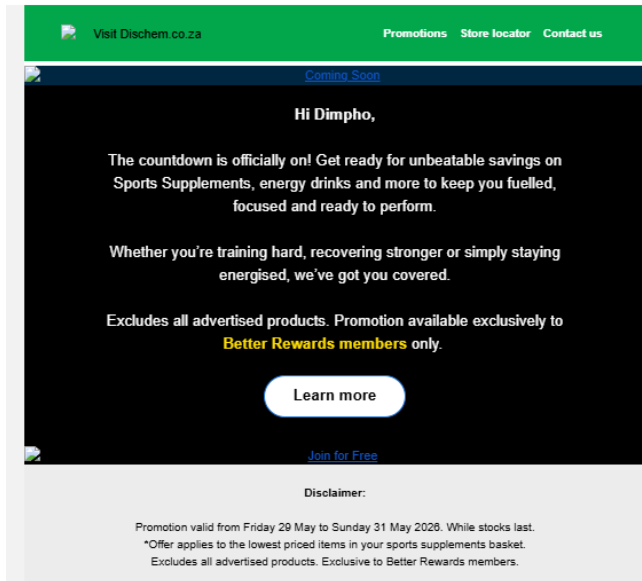
INTRODUCTION

In today's increasingly digital world, cyber security continues to evolve. Among the evolution, cyber threats continue to become more of a regular. These threats typically happen through phishing emails.

This report represents a real-world phishing example showcasing the email was sent from a fraudulent domain to the user. This analysis is conducted from the perspective of an analyst, examining real samples to understand how phishing attacks are structured.

PHISHING EMAIL ANALYSIS 1

At first glance this email is from the company “Dischem” informing the customer about an upcoming sale on sports supplements, energy drinks and many more. This can be believable to the receiver and they may be enticed by the sale.

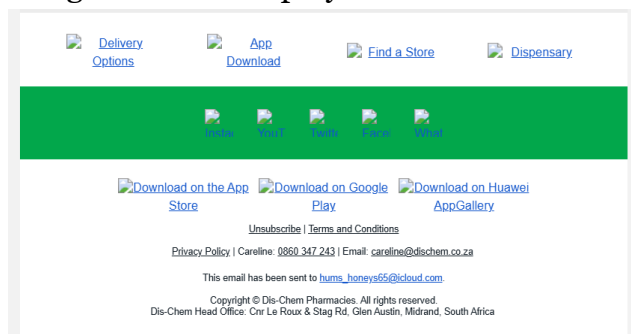


Indicators

- The sender's email uses “@icloud” instead of “@dischem”

From: Dis-Chem <comms_at_mail_dischem_co_za_s9fwc396f67gcd_32gt6433@icloud.com>

- The links in the email are buttons that redirect you to a fake login page
- Images are not displayed



HEADER ANALYSIS

Header Analyzed

Email Subject: The countdown to your next supplement haul starts NOW

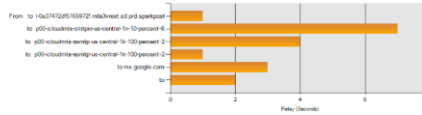
[Analyze New Header](#)

Delivery Information

- DMARC Compliant
- SPF Alignment
- SPF Authenticated
- DKIM Alignment
- DKIM Authenticated

Relay Information

Received Delay: 12 seconds



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	10.95.194.126	10.95.194.126	REST	5/25/2026 1:14:05 PM	✓
2	6 seconds	duneirmx.mail.dischem.co.za 137.22.240.39	p00-icloudmta-smtp-us-central-1n-100-percent-6	ESMTPS	5/25/2026 1:14:11 PM	✓
3	3 seconds	p00-icloudmta-smtp-us-central-1n-10-percent-6 p00-icloudmta-smtp-us-central-1n-100-percent-2	p00-icloudmta-smtp-us-central-1n-100-percent-2	ESMTPS	5/25/2026 1:14:14 PM	✓
4	0 seconds	outbound.ci.icloud.com 127.0.0.2	p00-icloudmta-smtp-us-central-1n-100-percent-2	ESMTPS	5/25/2026 1:14:14 PM	✓
5	2 seconds	outbound.ci.icloud.com 57.103.89.63	mx.google.com	ESMTPS	5/25/2026 1:14:16 PM	✓
6	1 Second		2002 a17-906 7803 b0 bbb db18 88ae	SMTP	5/25/2026 1:14:17 PM	✓

Findings

duneirmx.mail.dischem.co.za

137.22.240.39

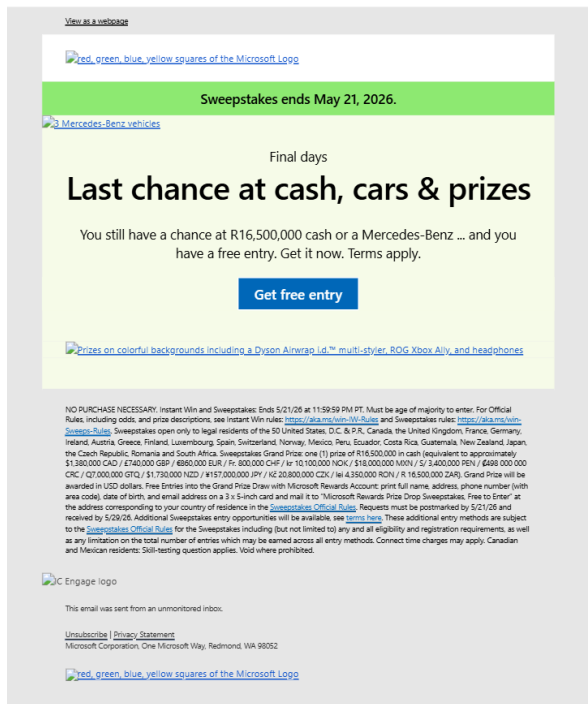
- It suggests that the message was originally sent from the Dis-Chem's mail infrastructure

Mx.google.com

This shows that the email uses iCloud's infrastructure to send the marketing email

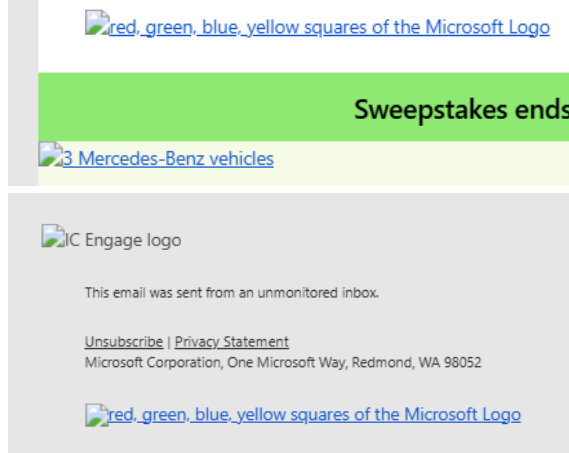
PHISHING EMAIL ANALYSIS 2

At first glance the email looks as though the receiver may stand a chance to win Microsoft rewards

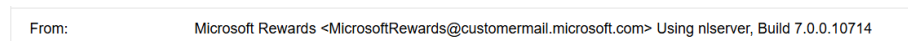


Indicators and risk

- The prizes are “too good to be true” all by a click of a button. These are used to lure people in
- Images are not loading



- Email looks realistic



HEADER ANALYSIS

Header Analyzed

Email Subject: You still have a free entry. Don't miss your shot at R16,500,000.

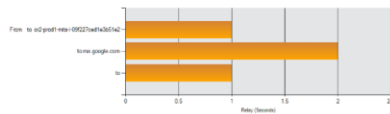
[Analyze New Header](#)

Delivery Information

- DMARC Compliant
- SPF Alignment
- SPF Authenticated
- DKIM Alignment
- DKIM Authenticated

Relay Information

Received Delay: 1 seconds



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	10.1.1.43	o2-prod1-mta-i-09027ced1e3b51a2-4.4.1.0	ESMTP	5/15/2026 5:26:20 PM	✓
2	1 Second	r12.customermail.microsoft.com 172.82.208.12	mx.google.com	ESMTPS	5/15/2026 5:26:21 PM	✗
3	0 seconds		2002:ac8:1415:0:b0:50e:62ab:d650	SMTP	5/15/2026 5:26:21 PM	

Findings

R12.customermail.microsoft.com

The email confirms the email was sent from IP address that is authorized by the domain owner

It shows us the email was digitally signed by the sender and not sent from somewhere else.

DMARC complaint

The DMARC is marked in red because Microsoft sends millions of marketing emails using “customermail.microsoft.com” email

PREVENTION & AWARENESS GUIDELINES

How to spot phishing email

- Offers that are “too good to be true”
- Email addresses that do not align with email that has been send out. Eg: An email from Netflix but the email address ends with “netflix.security.updater4592.com”
- Emails that have generic greeting instead of your actual name
- Images that do not load

Prevention

- Enforce multi-factor authentication, this adds an extra defense that stops them from logging in
- Use password manager as it will auto fill in your credentials but on a fake website/email, it will not auto fill
- Implement an email authentication protocol to prevent malicious emails
- Implementation of regular phishing simulations for employees

CONCLUSION

Phishing remains a persistent threat in cybersecurity. As demonstrated and shown in the document, attacks typically rely on manipulation, brand impersonation and tactics to trick unsuspecting the email receiver that will compromise their own security.